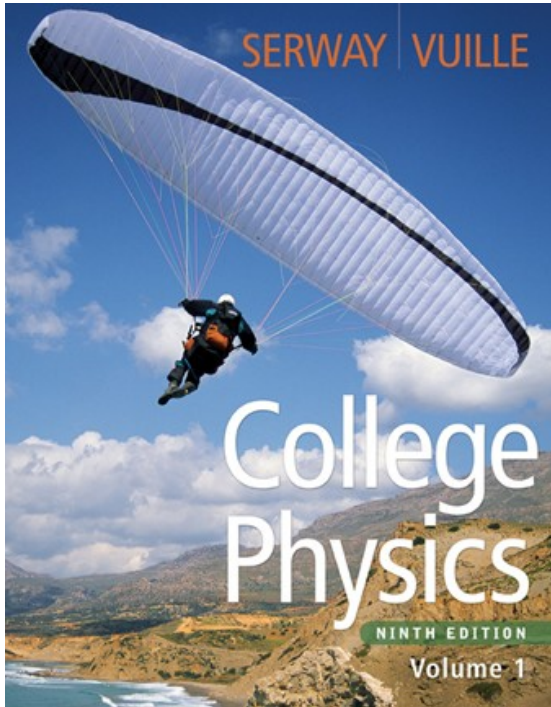




Raymond A. Serway
Chris Vuille

Chapter Seven

Rotational Motion and The Law of Gravity

Rotational Motion

- An important part of everyday life
 - Motion of the Earth
 - Rotating wheels
- Angular motion
 - Expressed in terms of
 - Angular speed
 - Angular acceleration
 - Centripetal acceleration

Gravity

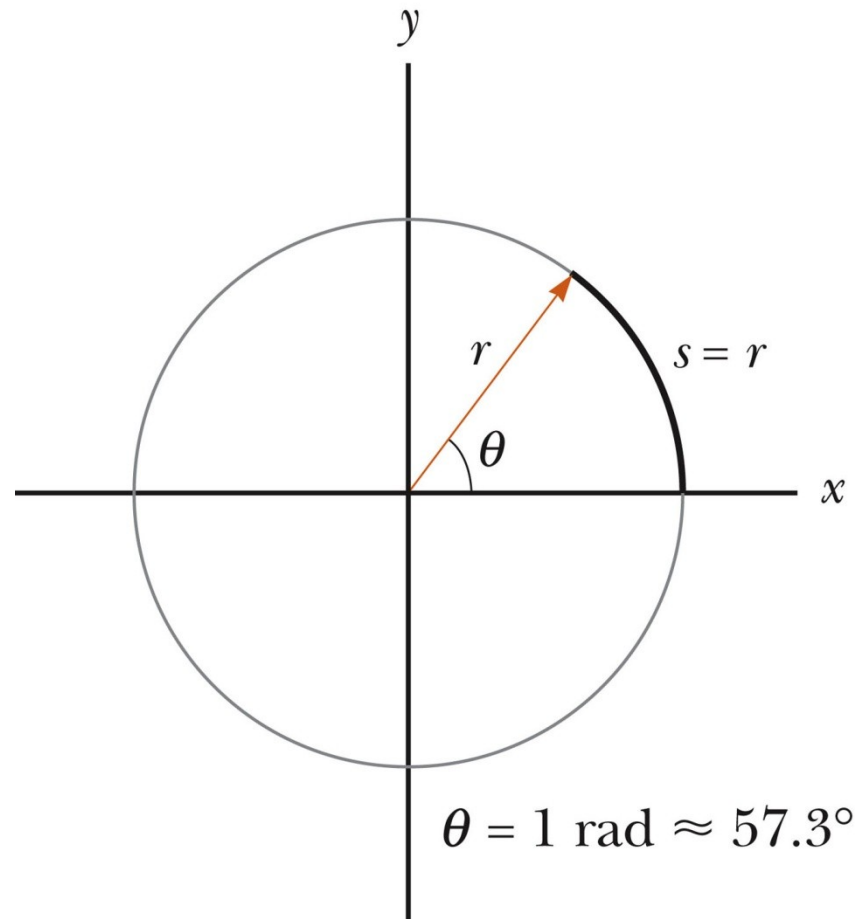
- Rotational motion combined with Newton's Law of Universal Gravity and Newton's Laws of motion can explain aspects of space travel and satellite motion
- Kepler's Three Laws of Planetary Motion
 - Formed the foundation of Newton's approach to gravity

Angular Motion

- Will be described in terms of
 - Angular displacement, $\Delta\theta$
 - Angular velocity, ω
 - Angular acceleration, α
- Analogous to the main concepts in linear motion

The Radian

- The radian is a unit of angular measure
- The radian can be defined as the arc length s along a circle divided by the radius r
- $\theta = \frac{s}{r}$



More About Radians

- Comparing degrees and radians

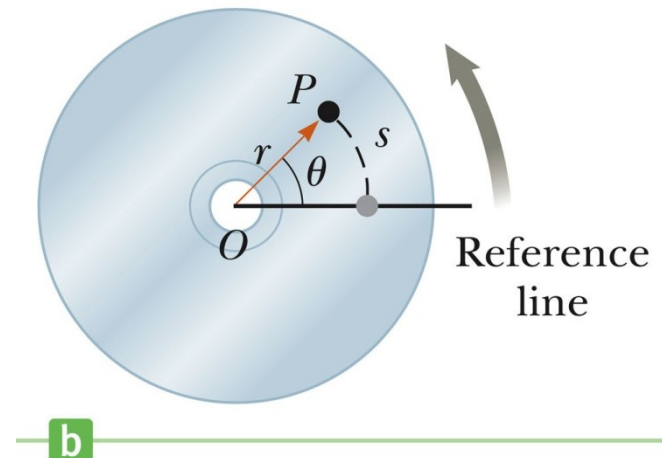
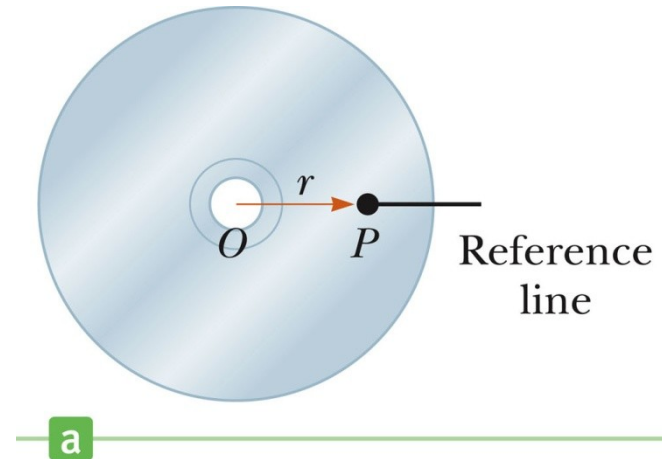
$$1 \text{ rad} = \frac{360^\circ}{2\pi} = 57.3^\circ$$

- Converting from degrees to radians

$$\theta [\text{rad}] = \frac{\pi}{180^\circ} \times \theta [\text{degrees}]$$

Angular Displacement

- Axis of rotation is the center of the disk
- Need a fixed reference line
- During time t , the reference line moves through angle θ
- The angle, θ , measured in radians, is the *angular position*

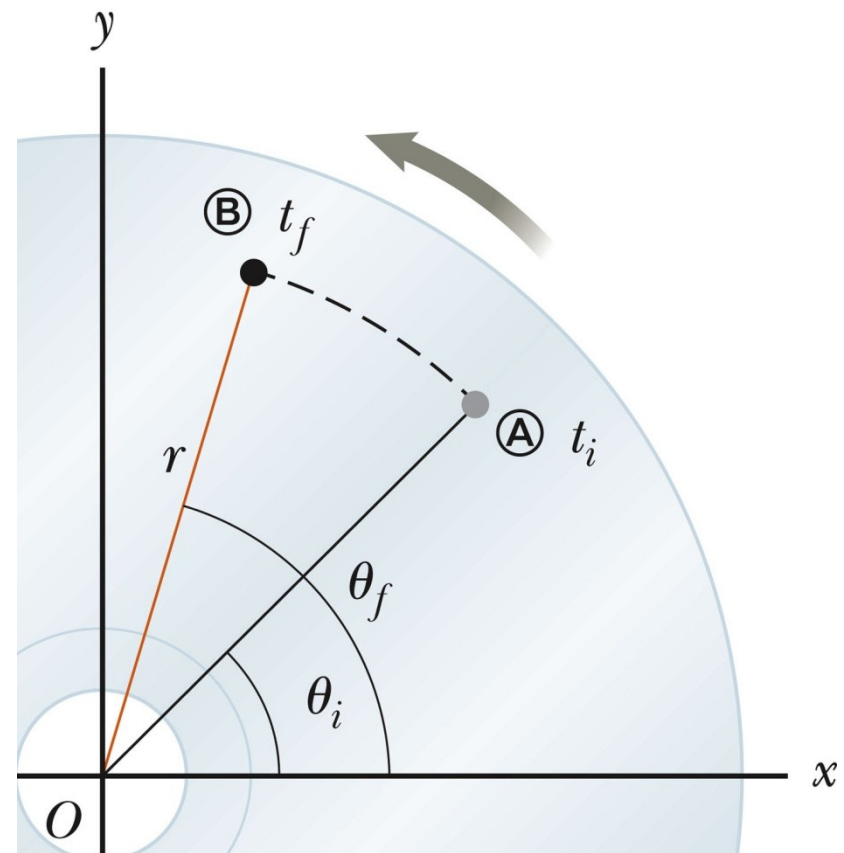


Rigid Body

- Every point on the object undergoes circular motion about the point O
- All parts of the object of the body rotate through the same angle during the same time
- The object is considered to be a **rigid body**
 - This means that each part of the body is fixed in position relative to all other parts of the body

Angular Displacement, cont.

- The *angular displacement* is defined as the angle the object rotates through during some time interval
- $\Delta\theta = \theta_f - \theta_i$
- The unit of angular displacement is the radian
- Each point on the object undergoes the same angular displacement



Average Angular Speed

- The average angular speed, ω , of a rotating rigid object is the ratio of the angular displacement to the time interval

$$\omega_{av} = \frac{\theta_f - \theta_i}{t_f - t_i} = \frac{\Delta\theta}{\Delta t}$$

Angular Speed, cont.

- The *instantaneous* angular speed is defined as the limit of the average speed as the time interval approaches zero
- SI unit: radians/sec
 - rad/s
- Speed will be positive if θ is increasing (counterclockwise)
- Speed will be negative if θ is decreasing (clockwise)
- When the angular speed is constant, the instantaneous angular speed is equal to the average angular speed

Average Angular Acceleration

- An object's average angular acceleration α_{av} during time interval Δt is the change in its angular speed $\Delta\omega$ divided by Δt :

$$\alpha_{av} = \frac{\omega_f - \omega_i}{t_f - t_i} = \frac{\Delta\omega}{\Delta t}$$

Angular Acceleration, cont

- SI unit: rad/s^2
- Positive angular accelerations are in the counterclockwise direction and negative accelerations are in the clockwise direction
- When a rigid object rotates about a fixed axis, every portion of the object has the same angular speed and the same angular acceleration
 - The tangential (linear) speed and acceleration will depend on the distance from a given point to the axis of rotation

Angular Acceleration, final

- The instantaneous angular acceleration is defined as the limit of the average acceleration as the time interval approaches zero

Analogies Between Linear and Rotational Motion

- There are many parallels between the motion equations for rotational motion and those for linear motion
- Every term in a given linear equation has a corresponding term in the analogous rotational equations

Linear Motion with a Constant
(Variables: x and v)

$$v = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2}at^2$$

$$v^2 = v_i^2 + 2a\Delta x$$

Rotational Motion About a Fixed Axis with α Constant (Variables: θ and ω)

$$\omega = \omega_i + \alpha t \quad [7.7]$$

$$\Delta \theta = \omega_i t + \frac{1}{2}\alpha t^2 \quad [7.8]$$

$$\omega^2 = \omega_i^2 + 2\alpha\Delta \theta \quad [7.9]$$

Relationship Between Angular and Linear Quantities

- Displacements

$$s = \theta r$$

- Speeds

$$v_t = \omega r$$

- Accelerations

$$a_t = \alpha r$$

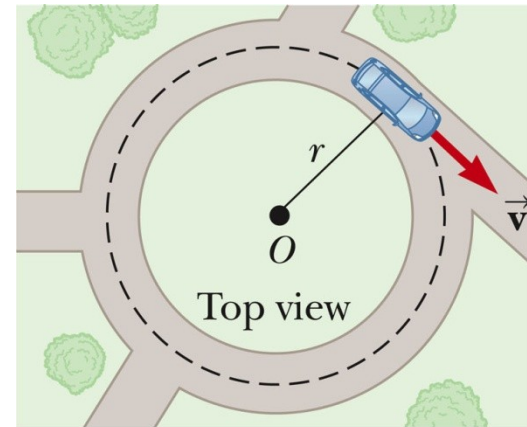
- Every point on the rotating object has the same angular motion
- Every point on the rotating object does *not* have the same linear motion

Centripetal Acceleration

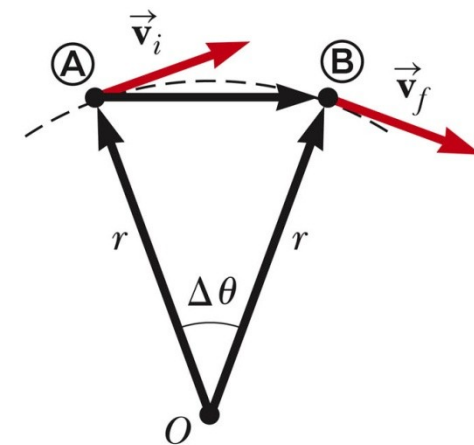
- An object traveling in a circle, even though it moves with a constant speed, will have an acceleration
- The centripetal acceleration is due to the change in the *direction* of the velocity

Centripetal Acceleration, cont.

- Centripetal refers to “center-seeking”
- The direction of the velocity changes
- The acceleration is directed toward the center of the circle of motion



a



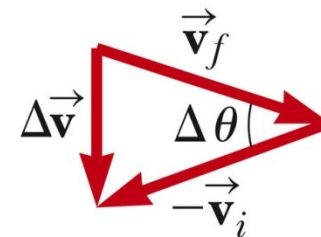
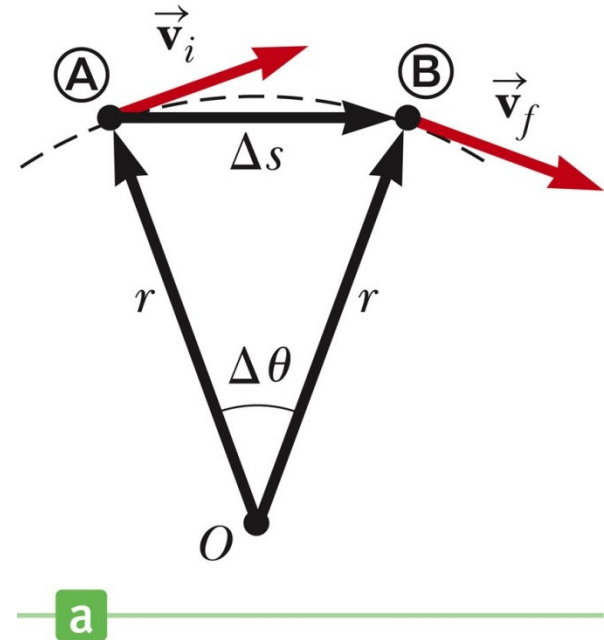
b

Centripetal Acceleration, final

- The magnitude of the centripetal acceleration is given by

$$a_c = \frac{v^2}{r}$$

- The direction is toward the center of the circle



Centripetal Acceleration and Angular Velocity

- The angular velocity and the linear velocity are related ($v = r \omega$)
- The centripetal acceleration can also be related to the angular velocity

$$a_c = \frac{v^2}{r} = \frac{r^2 \omega^2}{r} = r \omega^2$$

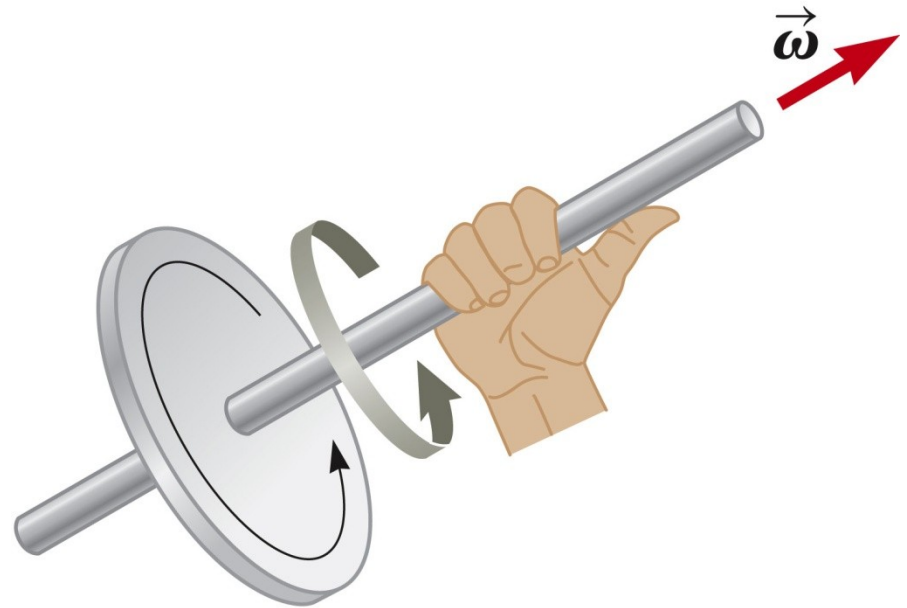
Total Acceleration

- The tangential component of the acceleration is due to changing speed
- The centripetal component of the acceleration is due to changing direction
- Total acceleration can be found from these components

$$a = \sqrt{a_t^2 + a_c^2}$$

Vector Nature of Angular Quantities

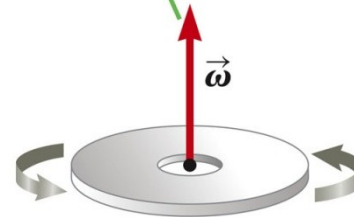
- Angular displacement, velocity and acceleration are all vector quantities
- Direction can be more completely defined by using the right hand rule
 - Grasp the axis of rotation with your right hand
 - Wrap your fingers in the direction of rotation
 - Your thumb points in the direction of $\vec{\omega}$



Velocity Directions, Example

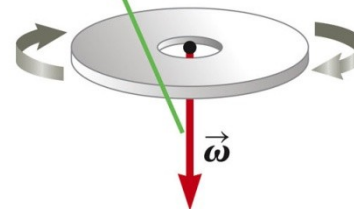
- In a, the disk rotates counterclockwise, the direction of the angular velocity is out of the page
- In b, the disk rotates clockwise, the direction of the angular velocity is into the page

When the disk rotates counterclockwise, $\vec{\omega}$ points upwards.



a

When the disk rotates clockwise, $\vec{\omega}$ points downwards.



b

Acceleration Directions

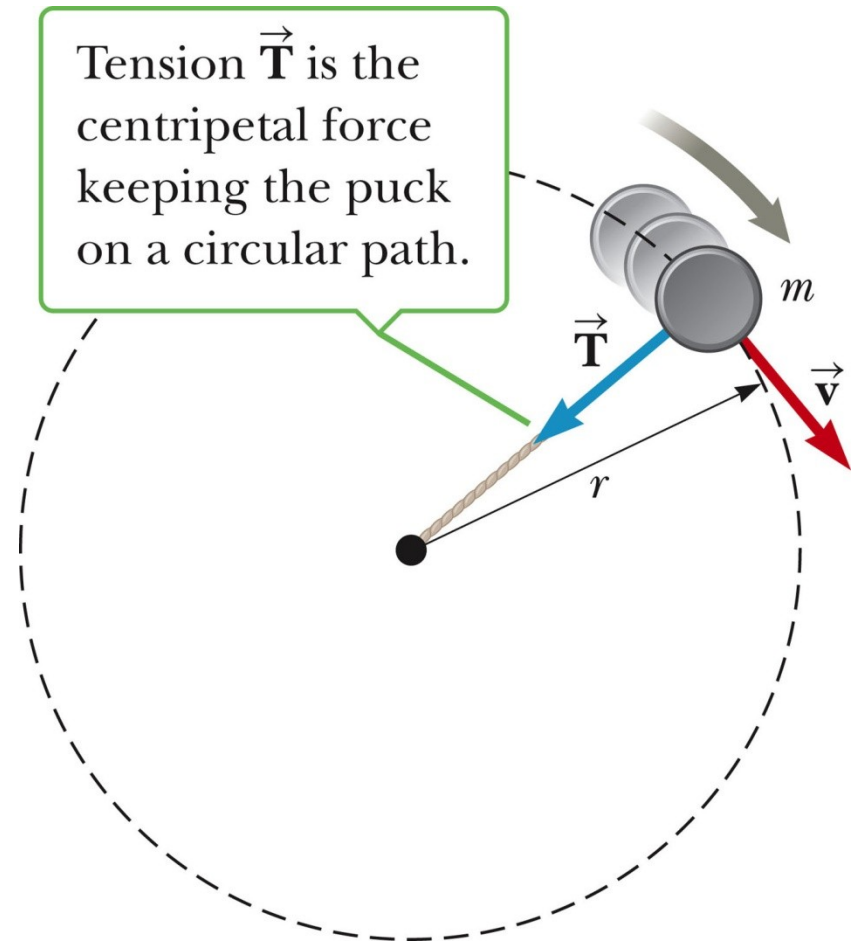
- If the angular acceleration and the angular velocity are in the same direction, the angular speed will increase with time
- If the angular acceleration and the angular velocity are in opposite directions, the angular speed will decrease with time

Forces Causing Centripetal Acceleration

- Newton's Second Law says that the centripetal acceleration is accompanied by a force
 - $F_c = ma_c$
 - F_c stands for any force that keeps an object following a circular path
 - Tension in a string
 - Gravity
 - Force of friction

Centripetal Force Example

- A puck of mass m is attached to a string
- Its weight is supported by a frictionless table
- The tension in the string causes the puck to move in a circle



Centripetal Force

- General equation

$$F_c = ma_c = \frac{mv^2}{r}$$

- If the force vanishes, the object will move in a straight line tangent to the circle of motion
- Centripetal force is a classification that includes forces acting toward a central point
 - It is *not* a force in itself
 - A centripetal force must be supplied by some actual, physical force

Problem Solving Strategy

- **Draw a free body diagram**, showing and labeling all the forces acting on the object(s)
- **Choose a coordinate system** that has one axis perpendicular to the circular path and the other axis tangent to the circular path
 - The normal to the plane of motion is also often needed

Problem Solving Strategy, cont.

- **Find the net force toward the center** of the circular path (this is the force that causes the centripetal acceleration, F_c)
 - The net radial force causes the centripetal acceleration
- **Use Newton's second law**
 - The directions will be radial, normal, and tangential
 - The acceleration in the radial direction will be the centripetal acceleration
- **Solve for the unknown(s)**

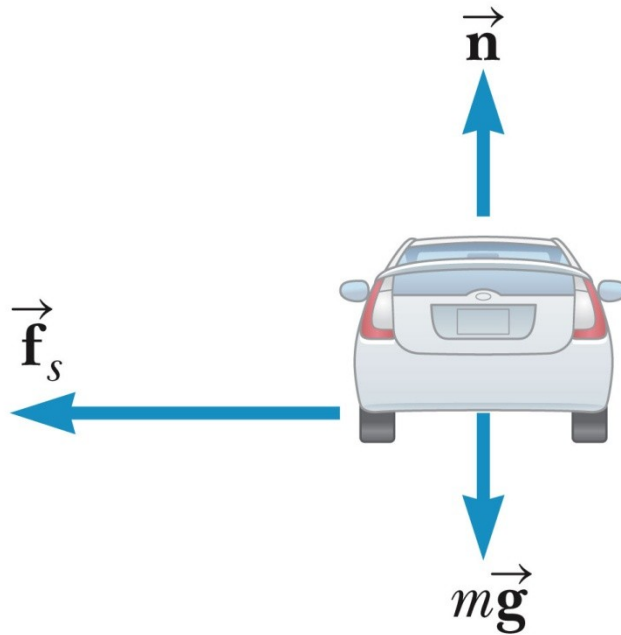
Applications of Forces Causing Centripetal Acceleration

- Many specific situations will use forces that cause centripetal acceleration
 - Level curves
 - Banked curves
 - Horizontal circles
 - Vertical circles

Level Curves

- Friction is the force that produces the centripetal acceleration
- Can find the frictional force, μ , or v

$$v = \sqrt{\mu rg}$$

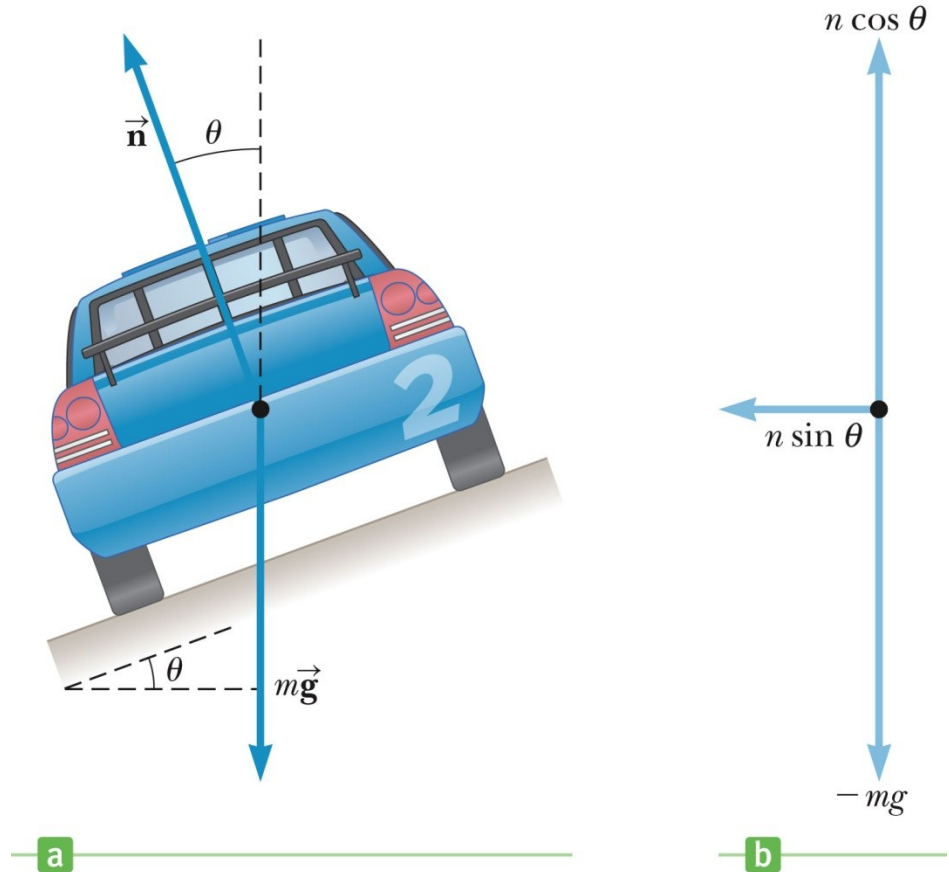


Banked Curves

- A component of the normal force adds to the frictional force to allow higher speeds

$$\tan\theta = \frac{v^2}{rg}$$

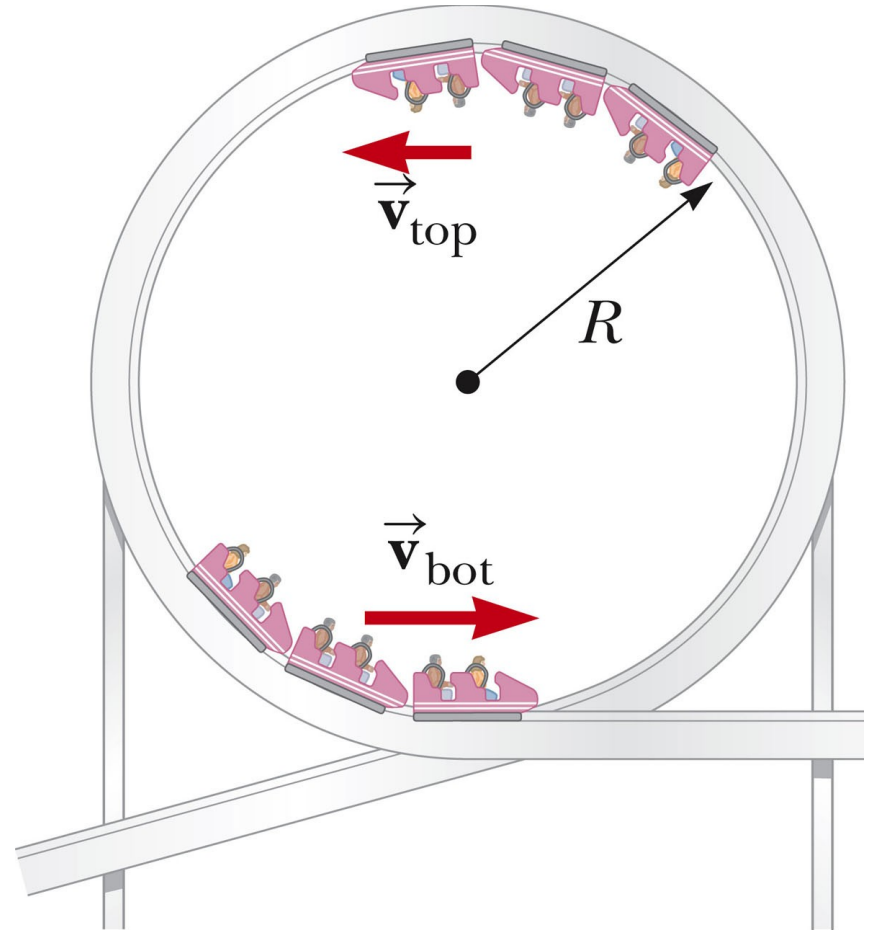
or $a_c = g \tan\theta$



Vertical Circle

- Look at the forces at the top of the circle
- The minimum speed at the top of the circle can be found

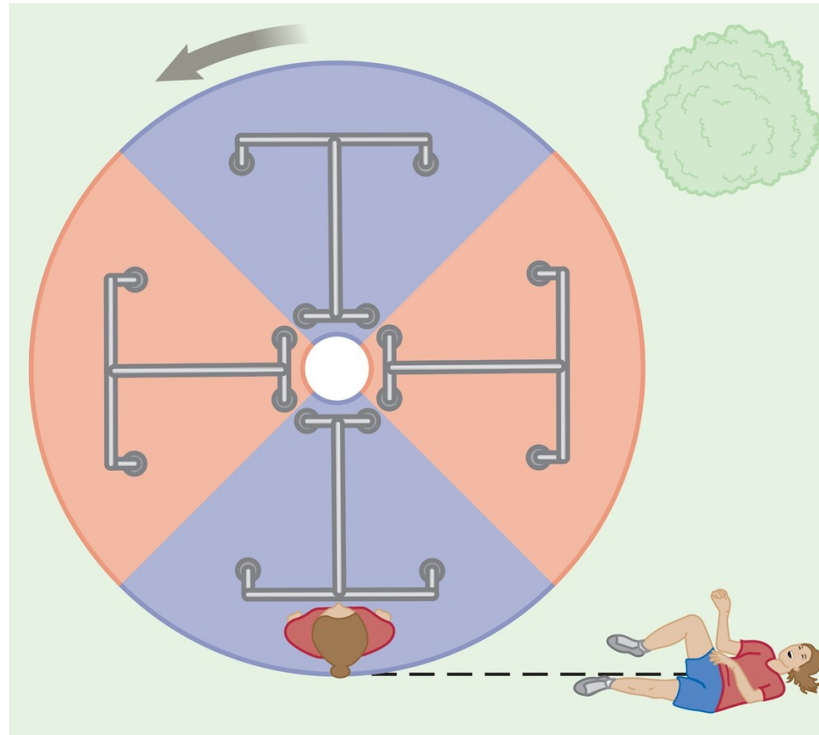
$$v_{\text{top}} = \sqrt{gR}$$



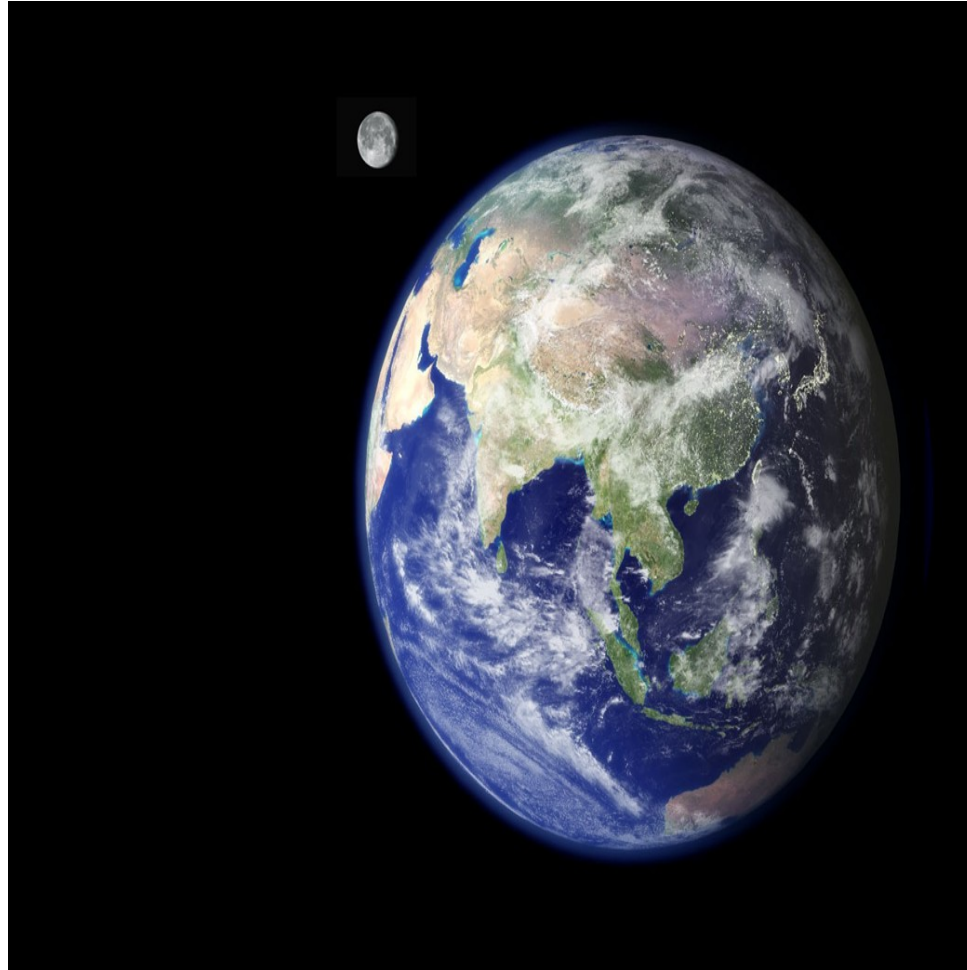
Forces in Accelerating Reference Frames

- Distinguish real forces from fictitious forces
- “Centrifugal” force is a fictitious force
 - It most often is the absence of an adequate centripetal force
 - Arises from measuring phenomena in a non-inertial reference frame

Fictitious Forces



Newtonian Gravitation (1 of 3)



Newton's Law of Universal Gravitation

- If two particles with masses m_1 and m_2 are separated by a distance r , then a gravitational force acts along a line joining them, with magnitude given by

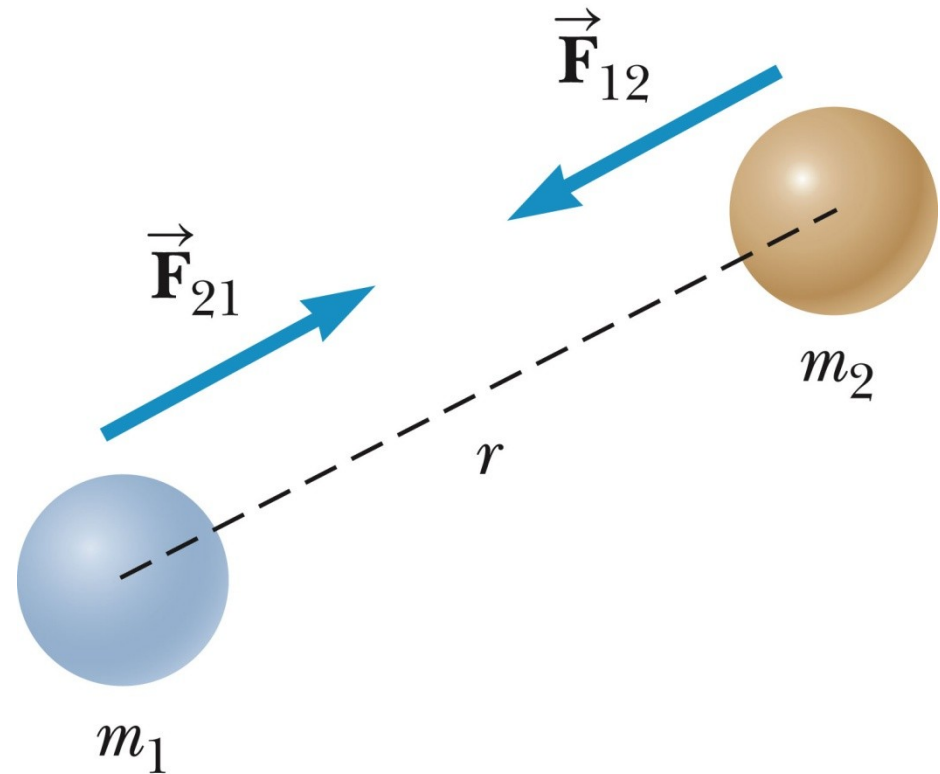
$$F = G \frac{m_1 m_2}{r^2}$$

Universal Gravitation, 2

- G is the constant of universal gravitational
- $G = 6.673 \times 10^{-11} \text{ N m}^2 / \text{kg}^2$
- This is an example of an *inverse square law*
- The gravitational force is always attractive

Universal Gravitation, 3

- The force that mass 1 exerts on mass 2 is equal and opposite to the force mass 2 exerts on mass 1
- The forces form a Newton's third law action-reaction

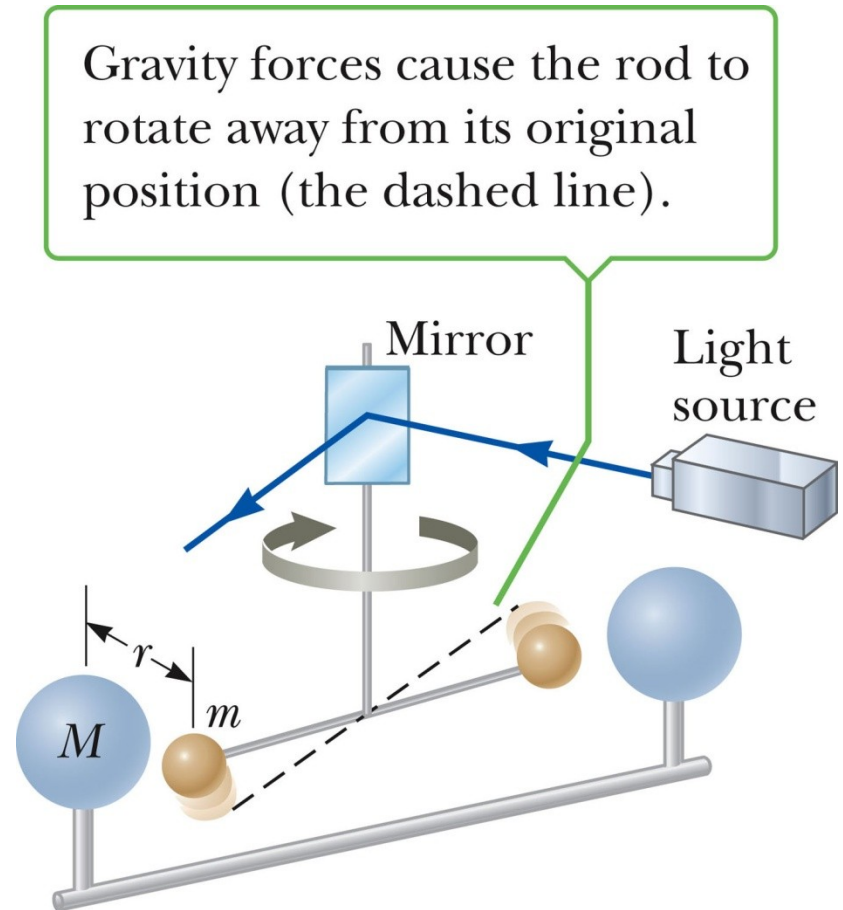


Universal Gravitation, 4

- The gravitational force exerted by a uniform sphere on a particle outside the sphere is the same as the force exerted if the entire mass of the sphere were concentrated on its center
 - This is called Gauss' Law

Gravitation Constant

- Determined experimentally
- Henry Cavendish
 - 1798
- The light beam and mirror serve to amplify the motion



Applications of Universal Gravitation

- Acceleration due to gravity
- g will vary with altitude

$$g = G \frac{M_E}{r^2}$$

- In general,

$$g_{\text{planet}} = G \frac{M_{\text{planet}}}{r^2}$$

Table 7.1 Free-Fall
Acceleration g at Various
Altitudes

Altitude (km) ^a	g (m/s ²)
1 000	7.33
2 000	5.68
3 000	4.53
4 000	3.70
5 000	3.08
6 000	2.60
7 000	2.23
8 000	1.93
9 000	1.69
10 000	1.49
50 000	0.13

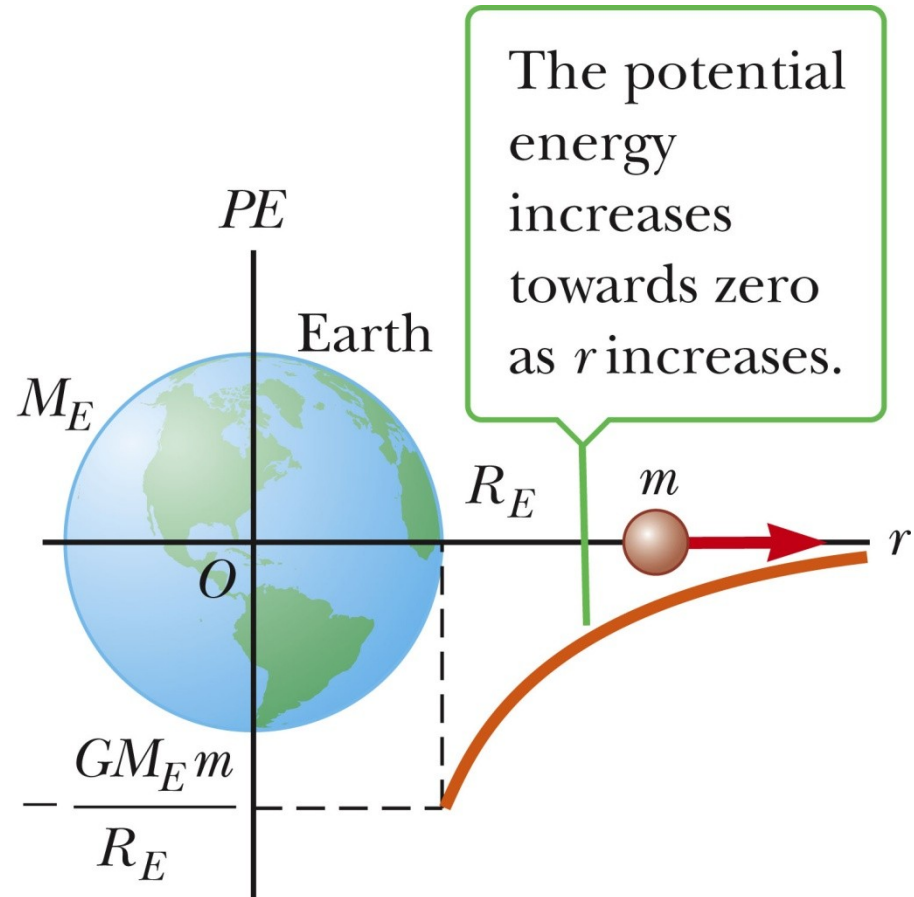
^aAll figures are distances above Earth's surface.

Gravitational Potential Energy

- $PE = mgh$ is valid only near the earth's surface
- For objects high above the earth's surface, an alternate expression is needed

$$PE = -G \frac{M_E m}{r}$$

- With $r > R_{\text{earth}}$
- Zero reference level is infinitely far from the earth



Escape Speed

- The escape speed is the speed needed for an object to soar off into space and not return

$$v_{\text{esc}} = \sqrt{\frac{2GM_E}{R_E}}$$

- For the earth, v_{esc} is about 11.2 km/s
- Note, v is independent of the mass of the object

Various Escape Speeds

- The escape speeds for various members of the solar system
- Escape speed is one factor that determines a planet's atmosphere

Table 7.2 Escape Speeds for the Planets and the Moon

Planet	v_{esc} (km/s)
Mercury	4.3
Venus	10.3
Earth	11.2
Moon	2.3
Mars	5.0
Jupiter	60.0
Saturn	36.0
Uranus	22.0
Neptune	24.0
Pluto ^a	1.1

^aIn August 2006, the International Astronomical Union adopted a definition of a planet that separates Pluto from the other eight planets. Pluto is now defined as a “dwarf planet” (like the asteroid Ceres).

Kepler's Laws

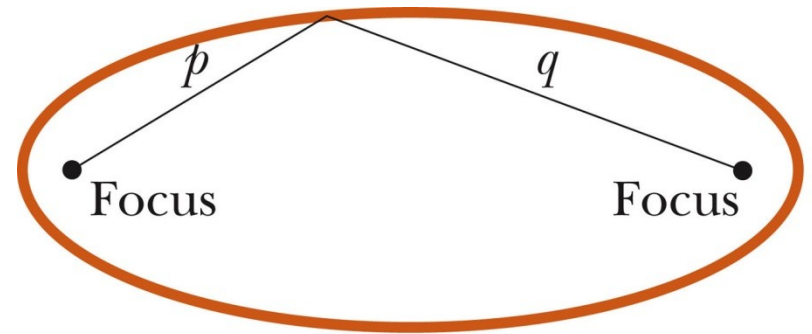
- All planets move in elliptical orbits with the Sun at one of the focal points.
- A line drawn from the Sun to any planet sweeps out equal areas in equal time intervals.
- The square of the orbital period of any planet is proportional to cube of the average distance from the Sun to the planet.

Kepler's Laws, cont.

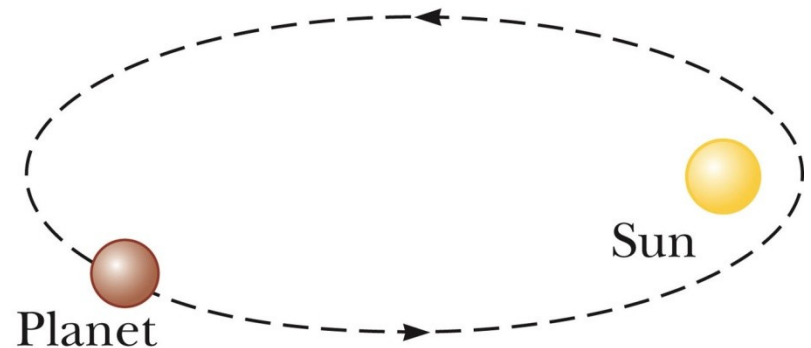
- Based on observations made by Brahe
- Newton later demonstrated that these laws were consequences of the gravitational force between any two objects together with Newton's laws of motion

Kepler's First Law

- All planets move in elliptical orbits with the Sun at one focus.
 - Any object bound to another by an inverse square law will move in an elliptical path
 - Second focus is empty



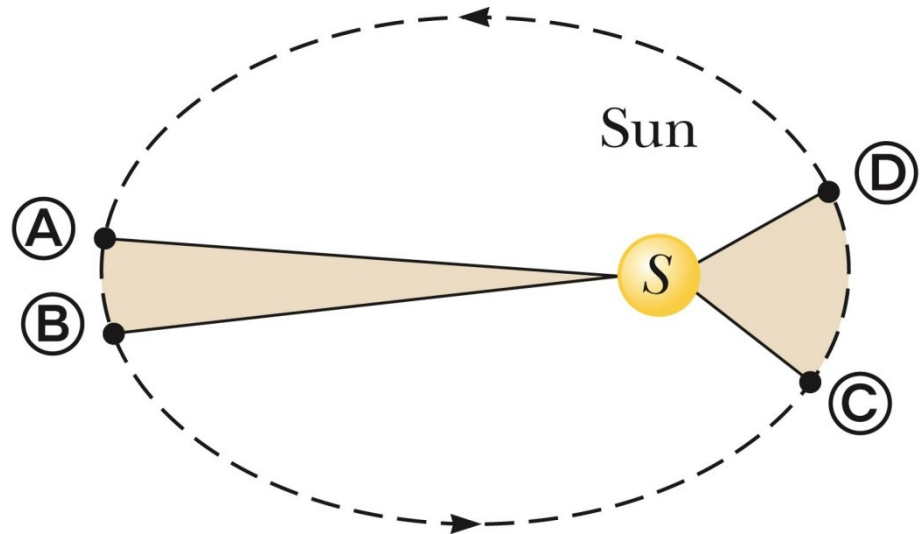
a



b

Kepler's Second Law

- A line drawn from the Sun to any planet will sweep out equal areas in equal times
 - Area from A to B and C to D are the same
 - The planet moves more slowly when farther from the Sun (A to B)
 - The planet moves more quickly when closest to the Sun (C to D)



Kepler's Third Law

- The square of the orbital period of any planet is proportional to cube of the average distance from the Sun to the planet.
 - T is the period (= time required for one revolution)
 - $T^2 = K a^3$
 - For orbit around the Sun, $K = K_s = 2.97 \times 10^{-19} \text{ s}^2/\text{m}^3$
 - K is independent of the mass of the planet

Kepler's Third Law (1 of 3)

$$M_p a_c = \frac{M_p v^2}{r} = \frac{GM_S M_p}{r^2}$$

$$v = \frac{2\pi r}{T} \rightarrow \frac{M_p \left(\frac{2\pi r}{T} \right)^2}{r} = \frac{GM_S M_p}{r^2}$$

$$T^2 = \left(\frac{4\pi^2}{GM_S} \right) r^3 = K_S r^3$$

$$K_S = \frac{4\pi^2}{GM_S} = 2.09 \times 10^{-19} \text{ s}^2/\text{m}^3$$

Kepler's Third Law (3 of 3)

Kepler's third law: $T^2 = a^3$, T in years and a in AU

Table 7.3 Useful Planetary Data

Body	Mass (kg)	Mean Radius (m)	Period (s)	Mean Distance from Sun (m)	$\frac{T^2}{r^3} 10^{-19} \left(\frac{s^2}{m^3} \right)$
Mercury	3.18×10^{23}	2.43×10^6	7.60×10^6	5.79×10^{10}	2.97
Venus	4.88×10^{24}	6.06×10^6	1.94×10^7	1.08×10^{11}	2.99
Earth	5.98×10^{24}	6.38×10^6	3.156×10^7	1.496×10^{11}	2.97
Mars	6.42×10^{23}	3.37×10^6	5.94×10^7	2.28×10^{11}	2.98
Jupiter	1.90×10^{27}	6.99×10^7	3.74×10^8	7.78×10^{11}	2.97
Saturn	5.68×10^{26}	5.85×10^7	9.35×10^8	1.43×10^{12}	2.99
Uranus	8.68×10^{25}	2.33×10^7	2.64×10^9	2.87×10^{12}	2.95
Neptune	1.03×10^{26}	2.21×10^7	5.22×10^9	4.50×10^{12}	2.99
Pluto ^a	1.27×10^{23}	1.14×10^6	7.82×10^9	5.91×10^{12}	2.96
Moon	7.36×10^{22}	1.74×10^6	—	—	—
Sun	1.991×10^{30}	6.96×10^8	—	—	—

^aIn August 2006, the International Astronomical Union adopted a definition of a planet that separates Pluto from the other eight planets. Pluto is now defined as a “dwarf planet” like the asteroid Ceres.

Kepler's Third Law, cont

- Can be used to find the mass of the Sun or a planet
- When the period is measured in Earth years and the semi-major axis is in AU, Kepler's Third Law has a simpler form
 - $T^2 = a^3$

Communications Satellite

- A geosynchronous orbit
 - Remains above the same place on the earth
 - The period of the satellite will be 24 hr
- $r = h + R_E$
- Still independent of the mass of the satellite